Task-Friendly Underwater Image Enhancement for Machine Vision Applications

Meng Yu, Liquan Shen, Zhengyong Wang, and Xia Hua

***Abstract*— Underwater images are often affected by color cast and blurring, which degrade the performance of underwater machine vision tasks. While existing underwater image enhancement (UIE) methods have been proposed to improve image quality for human perception, their effectiveness in enhancing machine vision performance is limited. In this article, a novel unsupervised UIE framework based on disentangled representation (DR) is proposed, which is designed for machine vision tasks. Specifically, the proposed framework disentangles the underwater image into two parts in the latent space according to whether they are beneficial to machine vision tasks: the task-friendly content features and the task-unfriendly distortion features. In addition, a semantic-aware contrastive module (SACM) is employed to alleviate the impact of losing key information required for machine vision tasks using the strategy of contrastive learning. Furthermore, two branches on the features and images are incorporated into the enhancement network, which serve the purpose of delivering task-relevant information to the enhancement model and guide the network to generate task-friendly images. Evaluation of the proposed method is conducted on multiple underwater image datasets, and a comparison is made with state-of-the-art enhancement methods in terms of machine vision performance. The experimental results demonstrate that the proposed method surpasses existing approaches in improving the accuracy and robustness of machine vision tasks, including object detection, semantic segmentation, and saliency detection in underwater environments. Our code is available at https://github.com/gemyumeng/TFUIE.**

***Index Terms*— Contrastive learning, disentangled representation (DR), machine vision tasks, underwater image enhancement (UIE).**

I. Introduction

UNDERWATER imaging is a crucial technique in various applications, such as marine biology, oceanography, and underwater exploration. However, when light propagates in water, it will be reflected, scattered, and absorbed, which makes the distribution of light in the underwater environment uneven, resulting in a dim image. In addition, particles, air bubbles, and water flow in the underwater environment will cause image noise and distortion. At the same time, as the depth increases, the scattering and absorption effects of light wavelengths will cause the image to appear blue or green, and the color distortion is more obvious. In addition, light rays are refracted and reflected as they travel through water, resulting in blurred and distorted images. These limitations seriously affect subsequent underwater applications, such as object detection [1], [2], [3] and semantic segmentation [4], [5].

To address this challenge, several underwater image enhancement (UIE) methods have been proposed. These methods can be classified into three categories: physics-based methods, image-based methods, and learning-based methods. Physics-based methods [6], [7], [8], [9], [10], [11], [12], [13], [14] consider the physical characteristics of underwater optical transmission and scattering to restore image details and colors by establishing optical transmission and scattering models. However, when the required model parameters cannot be accurately obtained or estimated, it can affect the effectiveness and reliability of the methods, indirectly limiting the development of underwater detection technologies. Image-based methods [15], [16], [17], [18], [19], [20], [21], [22], [23] primarily enhance the visual quality of images by modifying pixel values, such as histogram equalization, retinex decomposition, and image fusion. These methods may encounter limitations when dealing with complex image scenes and detail restoration, as they might not capture higher level semantic information. Learning-based methods [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34] employ deep learning models such as convolutional neural networks (CNNs) or generative adversarial networks (GANs) to enhance images by learning from a large amount of underwater image data. Deep learning methods typically require a substantial amount of annotated data for training. However, acquiring large-scale annotated data in the field of underwater imaging can be challenging and expensive. Furthermore, well-trained deep learning models may exhibit poor generalization capabilities in different underwater environments and conditions, resulting in degraded performance.

The mentioned methods primarily focus on optimizing perceptual quality based on human visual and do not consider subsequent machine vision tasks, as shown in Fig. 1(a)–(c), where the performance improvement of enhanced images in

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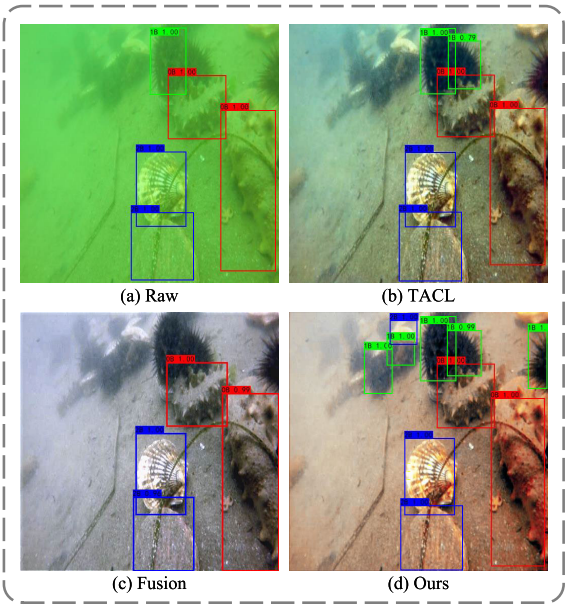


Fig. 1. Object detection results obtained by applying the raw images and various UIE methods to RetinaNet. (a) Results obtained by training the detector directly on the original images. (b)–(d) Results obtained using the TACL, Fusion, and our proposed method, respectively. (a)–(c) have a relatively serious missing detection situation, and our method effectively alleviates this problem and improves the detection performance.

object detection is limited. We have analyzed and summarized three reasons for this limitation.

1) *Inconsistent Data Characteristics:* UIE methods are typically trained or designed based on existing underwater image data. However, different underwater environments, lighting conditions, and water quality result in inconsistent data characteristics [24], [26]. When the characteristics of training data do not match the target task, the enhancement methods may not effectively adapt to the specific characteristics of the task data, thus providing limited performance improvement for machine tasks [35], [36].
2) *Information Loss and Distortion:* During UIE, certain image information may be lost or distorted due to noise removal, image smoothing, or contrast enhancement [37], [38]. This can lead to the loss or distortion of crucial information required by certain tasks, thereby affecting the accuracy and performance of machine tasks.
3) *Mismatched Enhancement Objectives:* UIE methods are typically designed to improve image quality and visibility, such as denoising, contrast enhancement, or color correction. However, performance improvement in machine tasks relies not only on the improvement of image quality but also on task-specific features and patterns. If the enhancement methods solely focus on image quality while neglecting task-related features, the enhanced results may not directly benefit machine tasks [36], [39], [40].

In this article, we propose a UIE method for machine vision tasks. Due to the lack of real underwater paired datasets, we employ an unsupervised enhancement approach using unpaired underwater real images and undistorted in-air images through cycle-consistent mapping. Furthermore, ordinary mappings may fail to accurately capture the intrinsic characteristics of the data and lack interpretability, making it difficult for trained models to generalize to task data with different data characteristics in complex underwater environments. Methods based on disentanglement can automatically learn the latent features and representations of images. These latent representations capture important information and semantic content in the images, exhibiting higher levels of abstraction and expressive power. This ability in feature and representation learning helps extract and utilize crucial features of the images, thereby improving the performance and effectiveness of image processing tasks. Therefore, we disentangle underwater images in the latent feature space into task-friendly content features and task-unfriendly distortion features. The content features are utilized in the subsequent enhancement steps, while the distortion features are combined with the content features of the in-air undistorted images to generate synthesized underwater images for unsupervised enhancement.

Typically, conventional UIE methods are designed based on human vision. However, during the enhancement process, some semantically useful details relevant to tasks may be lost due to the removal of underwater distortions. As a result, enhancing underwater images can actually decrease the accuracy of tasks such as object detection, semantic segmentation, and other related tasks. To address the issues of information loss and distortion, a semantic-aware feature extractor is employed to extract high-resolution semantic-aware content features that are independent of underwater distortions. Subsequently, the extracted features are combined into two triplets: positive sample pairs consisting of image features with the same semantic content and negative sample pairs consisting of image features with different semantic content. By utilizing a contrastive loss, the network is trained to preserve the details required for tasks as much as possible. This allows the network to learn how to retain important task-relevant details while minimizing distortion of task-irrelevant details.

UIE serves as a preprocessing step for machine vision tasks, but they are generally optimized as two separate tasks. When the enhancement methods solely focus on image quality while disregarding task-specific features, the enhanced results may not directly benefit machine tasks. To achieve closer integration between enhancement and machine vision tasks, we introduce task-guided branches from both feature and image perspectives. At the feature level, the underwater content encoder can be incentivized to extract more task-relevant semantic details. This encourages the extraction of features that are beneficial for the given machine vision task and helps improve the task's performance. At the image level, the enhanced image generator can be further guided to generate task-friendly images. Fig. 1 shows the object detection results of RetinaNet trained on the enhanced results of different enhancement algorithms, and our proposed method demonstrates a substantial improvement in detection performance.

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In our end-to-end network, each component is mutually constrained and jointly optimized with the overall objective of improving the performance of underwater machine vision tasks. To summarize, our contributions are given as follows.

1) We propose an unsupervised UIE framework toward machine vision tasks based on disentangled representation (DR), which is the first attempt to separate underwater images into two parts in the latent space according to whether they are beneficial to machine vision tasks: task-friendly content features and task-unfriendly distortion features.
2) We propose a semantic-aware contrastive module (SACM) that utilizes the strategy of contrastive learning to preserve the original semantic details of the underwater images as much as possible and alleviate the impact of losing key information required for machine vision tasks during the enhancement process.
3) To improve the performance of the enhancement on machine vision tasks, we introduce a dual task-guided branch (DTGB) on the enhancement network, which double-guides the enhancement toward task-friendly directions from the two levels of feature and image.

The rest of this article is organized as follows. Section II provides a review of related work in UIE. Section III presents our proposed method in detail. Section IV describes the experimental setup and presents the results of our method on various machine vision tasks. Finally, Section V concludes this article.

II. Related Work

A. UIE for Human Vision

UIE has been a subject of intense research in recent years due to the challenges posed by the complex and dynamic underwater environment. Various methods have been proposed to improve the quality and visibility of underwater images, including physics-based methods, image-based methods, and learning-based methods.

Physics-based methods are based on the underwater image formation model, where the background light and transmission map are estimated using certain prior assumptions. These priors include underwater dark channel prior, minimum information prior, blurriness prior, and so on. Berman et al. [6] incorporated color line priors and multispectral profile information of different water types into the physical model and leveraged the gray-world assumption theory for image dehazing. Peng and Cosman [7] proposed a method that combines blurriness to estimate more accurate scene depth in underwater image restoration. Inspired by the principle of minimum information loss, Li et al. [8] estimated an optimal transmission map to restore underwater images and utilize histogram distribution for contrast and brightness enhancement. Wang et al. [14] proposed a nonlocal attenuation-curve prior and used it to enhance contract and correct color for underwater images. These methods effectively restore underwater images in certain scenarios. However, the effectiveness and reliability of these methods can be affected when the priors are ineffective.

Image-based methods mainly improve the visual quality of the image by modifying the pixel value of the image, such as pixel stretching and adjustment, Retinex decomposition, and image fusion. Ghani and Isa [15] improved the visual quality of low-quality underwater images through a comprehensive approach of global and local contrast correction. Ancuti et al. [16] proposed a fusion-based strategy to enhance underwater images and videos, utilizing inputs and weight maps derived from the degraded image. After that, Ancuti et al. [17] blended two color-compensated and white-balanced images with weight maps to promote edge and color contrast transfer. Fu et al. [18] proposed a retinex-based method, which includes color correction, decomposing reflectance and illumination using a retinex variational framework and enhancing reflectance and illumination separately. However, most image-based methods cannot effectively remove chromatic aberration and improve contrast in complex underwater scenes.

Deep learning-based methods use deep learning models such as CNNs or GANs to enhance underwater images. Li et al. [24] employed a gated fusion network architecture to combine three input images obtained by applying white balance, histogram equalization, and gamma correction. Li et al. [25] utilized a multicolor space encoder network with an attention mechanism and a medium transmission-guided decoder network to address color casts and low contrast in underwater images. Islam et al. [26] presented a conditional GAN-based model for real-time UIE. Wang et al. [27] utilized a GAN to generate realistic underwater images from in-air images and depth maps, followed by color restoration and dehazing using U-Net trained on synthetic underwater data. Peng et al. [33] presented a U-shape Transformer network where the transformer model is for the first time introduced to the UIE task. Zhou et al. [34] utilized a guided color rendering image to direct the network's focus on color features in underwater images. Han et al. [29] designed a fast and novel model for comprehensive image understanding by cleverly capturing image information. However, the above three types of methods only consider the human visual perception, ignoring the requirements of machine vision tasks.

B. Task-Driven Image Enhancement

High-level tasks, such as object detection and semantic segmentation of in-air images, have achieved high performance, but in the presence of fog, rain, and noise, the performance of tasks will also drop significantly like underwater scenes.

In order to solve the problem of performance degradation of high-level tasks under extreme conditions, researchers try to combine the enhancement task and the subsequent high-level task, which are not regarded as two independent tasks. For example, Li et al. [41] proposed a lightweight dehazing network AOD-Net, which can be easily embedded into other deep models to improve high-level task performance, such as object detection on hazy images. Liu et al. [42] presented a CNN for image denoising that cascades two modules for image denoising and high-level tasks. Diamond et al. [43]

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utilized a novel end-to-end differentiable architecture for joint denoising, deblurring, and classification, which improves the accuracy of a classification network in low-light and challenging conditions. Sharma et al. [44] used a range of enhancement filters that can enhance image-specific details via end-to-end dynamic filter learning to improve image classification performance. Son et al. [45] proposed a universal and recognition-friendly image enhancement network, designed to enhance distorted images for improved performance of recognition models without retraining, which is capable of handling various image degradation factors and can be incorporated with any recognition model.

The task-driven enhancement methods mentioned above are challenging to apply to underwater images because it is difficult to obtain real reference images without distortion in underwater environments. Since the current approaches that consider both aspects of human vision and machine vision take an unsupervised learning approach, Chen et al. [46] proposed two perceptual enhancement models that use a deep enhancement model with a detection perceptor, providing coherent information in the form of gradients to generate visually pleasing images or detection favorable images. Liu et al. [47] improved the accuracy of object detection in underwater images by using a bilateral constrained closed-loop adversarial enhancement module, contrastive cues, and task-aware feedback to achieve both visually-friendly and detection-oriented enhancement, which was validated by using a series of detectors and semantic segmentation. However, their performance in tasks, such as object detection, is worse compared to the raw images.

C. Disentangled Representation

DR has the potential to improve the interpretability, generalization, and robustness of machine learning models, which have been successfully applied in various image processing tasks such as style transfer, denoising, and dehazing [48], [49], [50], [51], [52], [53]. By learning a set of independent and interpretable factors of variation in the input data, DR can improve the quality, efficiency, and flexibility of image processing algorithms.

Lee et al. [48] embedded images into domain-invariant content space and domain-specific attribute space based on DR for image-to-image translation. Lu et al. [49] achieved domain-specific, single-image unsupervised deblurring by splitting content and blur features using encoders and enforcing a Kullback–Leibler (KL) divergence loss to regularize blur attributes. Li et al. [50] proposed an unsupervised single image dehazing network based on DR and a compact multiscale feature attention module, achieving competitive dehazing results and relatively high processing speed. Tang et al. [51] proposed a user-guided framework for low-light image enhancement that models images as a combination of content and exposure attributes, disentangles them using content and attribute encoders, and reconstructs enhanced images by combining the scene content of low-light images with the exposure attributes of guide images through a generator. Despite the potential benefits of DR, its utilization in underwater image processing has been scarce.

III. Proposed Methods

A. Overview of the Proposed Framework

General UIE methods focus on the improvement of human perception quality while ignoring the impact of enhancement on machine vision tasks. In fact, the accuracy of these tasks using the enhanced results may be reduced compared to the original, unprocessed images. Therefore, we propose a task-friendly UIE method aimed at improving the performance of machine vision tasks. Due to the lack of paired real underwater datasets, we use an unsupervised method based on DR to enhance underwater images, as shown in Fig. 2. To ensure the preservation of semantic details during the image transformation process, we present an SACM within the framework. This module extracts high-resolution features that contain rich semantic details and enforces their correlation with the content of the image, independent of distortions, through a contrastive loss. This constraint enables the enhanced image to retain valuable underwater-specific details. In addition, we incorporate task branches for the extracted features and the enhanced image separately, introducing two task losses that further constrain the enhancement network to generate more task-friendly images.

Specifically, given an underwater image X and an in-air image Y, X gets an enhanced image Y_X through an enhancement network, and Y gets a synthesized underwater image X_Y through a synthesis network. The enhanced image and synthesized image are sent to the enhancement and synthesis network to obtain the reconstructed underwater image $\hat{X}$ and in-air image $\hat{Y}$, forming a cycle-consistency network. To increase the interpretability of the network, we map unpaired images to feature space with two different content encoders (E_U^C and E_A^C) and a distortion encoder (E_U^D). The content encoder is used to extract the task-friendly content features, and the distortion encoder is used to extract the task-unfriendly distortion features. The content features of the underwater image are fed into the enhanced image generator (G_{enh}), while the distortion features combined with the content features of the in-air image are fed into the synthesized image generator (G_{syn}).

B. Disentanglement of Content and Distortion

The datasets in different underwater environments can exhibit diverse data characteristics, making it challenging for UIE methods to effectively generalize to different task-specific data. In order to improve the network's generalization capability and interpretability, we incorporate DR into our framework, as shown in Fig. 2. For images in the underwater domain, we employ an underwater content encoder E_U^C and an underwater distortion encoder E_U^D to disentangle the task-friendly content features and the task-unfriendly distortion features. The underwater content encoder is responsible for extracting the content features specific to underwater images, while the underwater distortion encoder captures the distortion-related features caused by underwater conditions. For images in the in-air domain, only an in-air content encoder E_A^C is required to extract their content features. Once the features are extracted, an enhanced image generator G_{enh} and a synthetic image generator G_{syn} are utilized to generate RGB images separately

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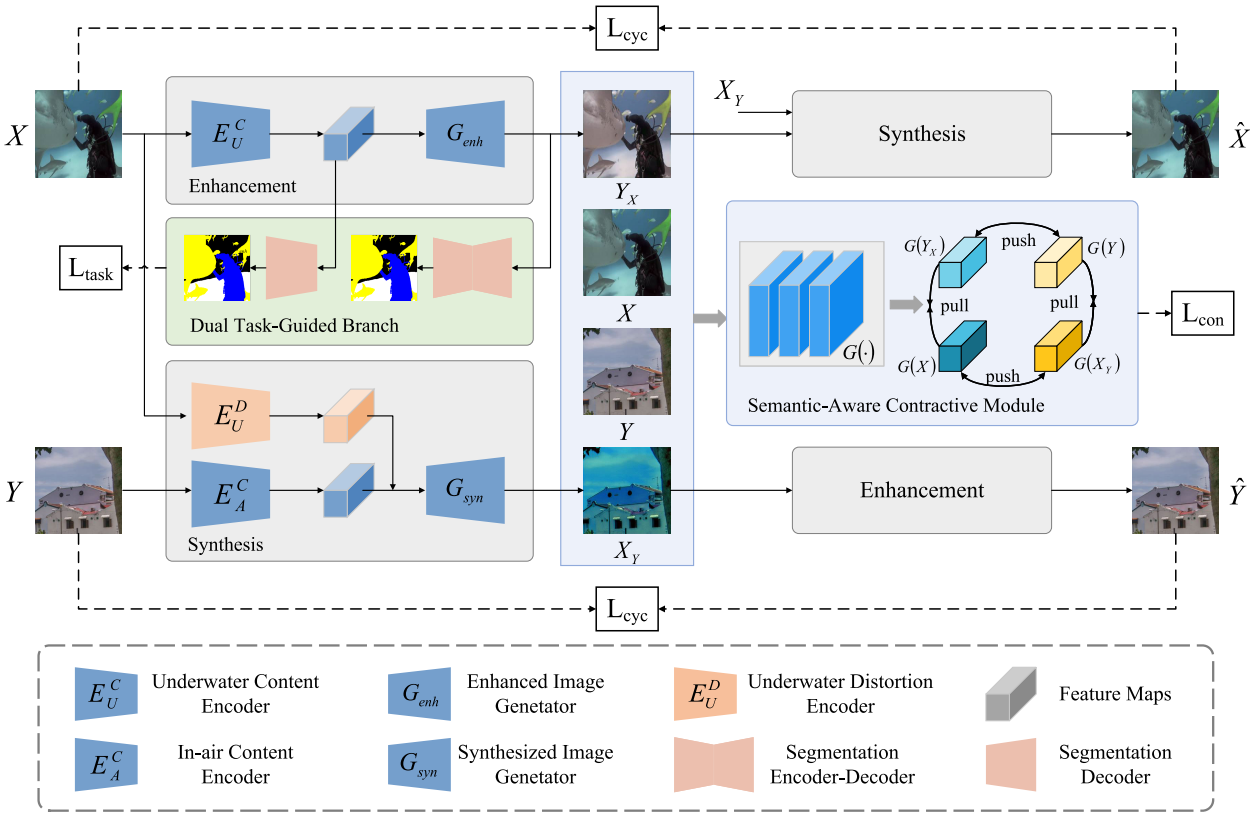


Fig. 2. Overview of the proposed method. It consists of two enhancement and synthesis processes based on DR, an SACM, and a DTGB.

for each domain. The framework consists of two stages: forward cross translation and backward cross translation.

1) Forward Cross Translation: Given an underwater image X and an in-air image Y, the process of enhancement and synthesis is represented by

$$Y_X = G_{\text{enh}}\left(E_U^C(X)\right) \tag{1}$$

$$X_Y = G_{\text{syn}}\left(E_U^D(X) + E_A^C(Y)\right) \tag{2}$$

where Y_X represents the enhanced image, X_Y denotes the synthesized image, and G_{enh} and G_{syn} are viewed as specific domain generators.

2) Backward Cross Translation: After performing the forward cross translation, the reconstructed image can be achieved by swapping the inputs X_Y and Y_X that

$$\hat{Y} = G_{\text{enh}}\left(E_U^C(X_Y)\right) \tag{3}$$

$$\hat{X} = G_{\text{syn}}\left(E_U^D(X_Y) + E_A^C(Y_X)\right) \tag{4}$$

where $\hat{X}$ and $\hat{Y}$ represent the reconstructed images of the original underwater images X and the original in-air image Y, respectively.

The objective of forward cross translation is to generate enhanced images Y_X and synthesized images X_Y, while the purpose of backward cross translation is to utilize the results of the forward translation and the raw images as constraints. The network details are shown in Fig. 3. We employ the encoder–decoder structure based on U-Net and replace the ordinary convolution with dense block. The content encoder and distortion encoder share the same network structure. After feeding the input image into the encoder, it undergoes four downsampling stages, resulting in content and distortion features at four different scales. These features are then separately fed into two generators, each with the same structure. It is

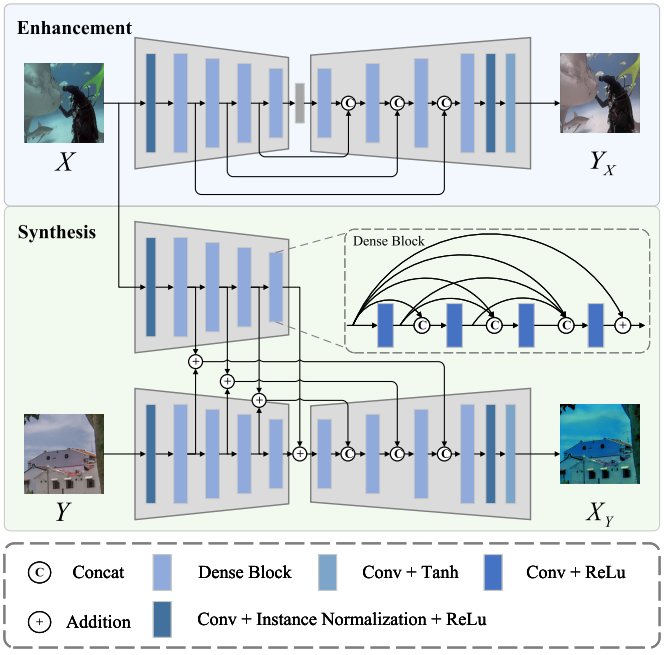


Fig. 3. Structure of enhancement and synthesis network based on DR, where three encoders and two generators operate independently with nonshared weights.

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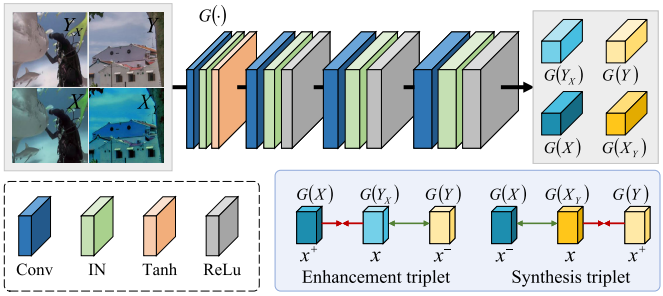


Fig. 4. Structure of the SACM. $G(\cdot)$ represents a feature extractor, which is used to extract semantic-aware features. The extracted features form two triplets, x represents the anchor, x^+ represents the positive sample, and x^- represents the negative sample.

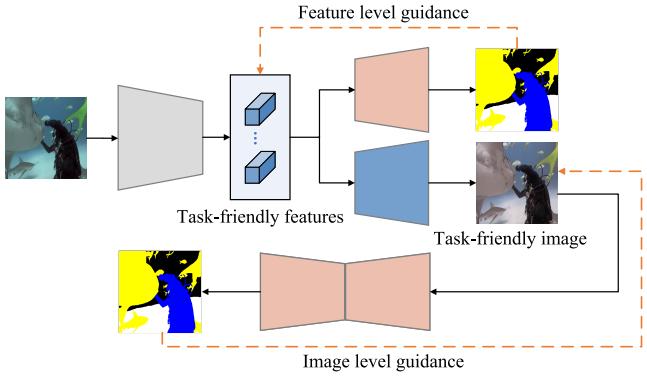


Fig. 5. Structure of the DTGB.

worth noting that although the network structures of the individual encoders and decoders are identical, their weights are not shared. The dense connections in the encoder–decoder architecture allow for efficient information flow and gradient propagation, facilitating the learning of complex relationships between the underwater and in-air domains. The multiscale fusion strategy enables the model to capture both local and global contextual information, enhancing the ability to disentangle the underwater distortion and incorporate the in-air content effectively.

C. Semantic-Aware Contrastive Module

When performing UIE, common processing methods include noise removal, image smoothing, and contrast enhancement. These operations may introduce certain errors or perform local image repairs, thereby altering the characteristics of the original image. While these processes can improve the visual quality of underwater images, they may result in the loss of some fine detail information that is relevant to machine vision tasks. In order to minimize the loss of detail information, we introduce the principle of contrastive learning into our framework. Specifically, we design an SACM that leverages contrastive loss to enable the semantic-aware feature extractor to extract semantic-aware content features. This ensures that the enhanced underwater images, while mitigating the distortions caused by underwater conditions, preserve their original structural details.

The structure of SACM is shown in Fig. 4. $G(\cdot)$ represents the semantic-aware feature extractor, which is used to extract the features of original underwater image X, in-air image Y, enhanced image Y_X, and synthesized image X_Y, denoted as $G(X)$, $G(Y)$, $G(Y_X)$, and $G(X_Y)$. The extracted features have the same resolution as the input image to preserve rich detail information. We employ a contrastive learning strategy to make the enhanced image retain as much detail as possible in the original image. These features form two triplets. In the enhancement triplet, the feature $G(Y_X)$ of the enhanced image serves as the anchor, while the feature $G(X)$ of the underwater image with the same content and structure acts as the positive sample, and $G(Y)$ serves as the negative sample. In the synthesis triplet, the anchor is the feature $G(X_Y)$ of the synthesized image, the positive sample is $G(Y)$, and the negative sample is $G(X)$. By pushing the anchor closer to its positive sample with the same content and pulling it further away from the negative sample with different content, we ensure that the enhancement process does not discard potentially beneficial information for machine vision tasks that are inherent in the original image.

D. Dual Task-Guided Branch

UIE plays a crucial role as a preprocessing step in underwater machine vision tasks. However, traditionally, these two processes have been optimized separately, lacking a close integration. In order to establish a stronger connection between enhancement and machine vision tasks, we introduce task-guided branches at both feature and image levels, thereby addressing this limitation. At the feature level, our approach encourages the underwater content encoder to extract semantic details that are specifically relevant to tasks. By incorporating task-specific incentives, the encoder becomes more adept at capturing features that are essential for subsequent machine vision tasks. Moving to the image level, our framework further guides the enhanced image generator to produce three-channel images that are beneficial for the tasks. This is achieved by considering the task-specific requirements during the enhancement process, thereby creating images that better support subsequent machine vision tasks.

To realize these objectives, we employ a multiscale content feature extraction mechanism from the input underwater image. As shown in Fig. 5, features extracted by the underwater content encoder $E_U^C(X)$ are sent to two networks. One is directed to a decoder for predicting the result of the guidance task, which is the feature-level task-guided branch guiding the underwater content encoder to extract task-friendly features. The other is initially directed to the enhanced image generator, and the generated enhanced images Y_X are subsequently input into a task network. This represents the image-level task-guided branch, which guides the enhancement network to generate task-friendly images. This dual-task approach facilitates a comprehensive exploration of the underlying features in both the original and enhanced images, contributing to more accurate and reliable results. By incorporating task-guided branches at both the feature and image levels, our proposed framework effectively constrains the network to prioritize and emphasize features that are specifically advantageous for underwater machine vision tasks.

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E. Loss Function

In this section, we give a detailed description about the loss functions. We apply cycle-consistency loss, identity mapping loss, adversarial loss, contrastive loss, and task loss to train the network in the proposed framework.

1) Cycle-Consistency Loss: Due to the presence of two closed loops in our framework, we incorporate a cycle-consistency loss to ensure the smooth progression of information during both the enhancement and synthesis processes. This loss function serves as a supervisory signal, guiding the enhancement and synthesis networks to retain the content of the input images. We define the cycle-consistency loss as follows:

$$L_{\text{cc}} = \left\|\hat{X} - X\right\|_1 + \left\|\hat{Y} - Y\right\|_1. \tag{5}$$

2) Identity Mapping Loss: The identity mapping loss is applied to encourage feature preservation, improve the quality of translated images, and stabilize the training process. The identity mapping loss is defined as

$$L_{\text{idt}} = \left\|G_{\text{syn}}\left(E_U^D(Y) + E_A^C(Y)\right) - Y\right\|_1. \tag{6}$$

3) Adversarial Loss: The proposed network uses adversarial loss functions L_{adv}, where D represents an attempt to discriminate real underwater images X and synthesized underwater images X_Y. The adversarial loss is defined as

$$L_{\text{adv}} = -\log D(X) - \log(1 - D(X_Y)). \tag{7}$$

4) Contrastive Loss: To fully preserve task-related content structure information, we extract semantic-aware features with a feature extractor $G(\cdot)$ and employ $L1$ regularization to strengthen the contrastive prior, which is expressed as

$$L_{\text{con}} = \frac{\left\|G(Y_X) - G(X)\right\|_1}{\left\|G(Y_X) - G(Y)\right\|_1} + \frac{\left\|G(X_Y) - G(Y)\right\|_1}{\left\|G(X_Y) - G(X)\right\|_1}. \tag{8}$$

5) Task Loss: In the DTGB, we use the underwater semantic segmentation task to train jointly with the enhancement network. The task loss consists of two parts, constrained on features and images, defined as

$$L_{\text{task}} = \text{CE}\left(D_{\text{seg}}\left(E_U^C(X)\right), \text{mask}\right) + \text{CE}\left(N_{\text{seg}}(X_Y), \text{mask}\right) \tag{9}$$

where $\text{CE}(\cdot)$ denotes the cross-entropy loss for semantic task, D_{seg} denotes a decoder for segmenting the task-friendly content features, and N_{seg} denotes a U-Net network for segmenting enhanced images. Mask represents the semantic label of the training data.

6) Overall Loss Function: For the joint training, the total loss function is defined as

$$L_{\text{total}} = \lambda_1 L_{\text{cc}} + \lambda_2 L_{\text{idt}} + \lambda_3 L_{\text{adv}} + \lambda_4 L_{\text{con}} + \lambda_5 L_{\text{task}} \tag{10}$$

where λ_1–λ_5 are hyperparameters, which are set to 10, 5, 1, 5, and 0.2, respectively.

IV. Experiments

A. Datasets and Metrics

Due to the lack of paired training data, we trained our model using the underwater dataset SUIM [4] and the in-air dataset Flickr2K [55]. The SUIM dataset consists of 1634 underwater images with semantic labels, so we randomly selected 1634 images from the Flickr2K dataset. We evaluate our method on the performance of underwater machine vision tasks on object detection, semantic segmentation, and saliency detection. The ChinaMM [56] dataset is utilized for object detection, which comprises 2071 training images and 676 testing images. The dataset includes three categories of objects: holothurian, echinus, and scallop. For semantic segmentation, we employed the SUIM dataset, which labels underwater images into eight categories (background waterbody, human divers, plants/seagrass, wrecks/ruins, robots/instruments, reefs/invertebrates, fish/vertebrates, and sea-floor/rocks). As for saliency detection, we used the UFO-120 [57] dataset and the USOD10K dataset [58]. The UFO-120 dataset consists of 1500 training images and 120 testing images with the corresponding saliency labels. The USOD10K dataset consists of 10 255 underwater images, covering 70 categories of salient objects in 12 different underwater scenes.

Our enhancement methods are evaluated solely for machine vision tasks, with considerations regarding the visual quality perceived by human observers being excluded. For object detection, we employ the mAP@0.5:0.95 metric to quantify the average precision across various object categories. Regarding semantic segmentation, we employ mean intersection over union (mIoU) and mean pixel accuracy (mPA) as assessment metric for evaluating the performance of the segmentation. mIoU is a widely adopted evaluation measure that captures the extent of overlap between the predicted results and the ground truth. Concerning saliency detection, we utilize F-measure, E-measure, S-measure, and mean absolute error (MAE) to comprehensively evaluate the performance of salient object detection. F-measure is computed as the weighted harmonic mean of precision and recall. E-measure quantifies saliency error by considering the Euclidean distance between predicted saliency regions and the corresponding ground truth regions. S-measure assesses the structural similarity between the detection map and the ground truth. Finally, MAE measures the average absolute discrepancy of pixel values between the detection map and the ground truth.

B. Implementation Details

All the experiments are implemented using PyTorch and trained on a Linux workstation with NVIDIA GeForce RTX 2070. We adopted PatchGAN [59] to construct the discriminator and chose underwater semantic segmentation as the guidance task. During the training process, we randomly cropped 1634 images with a resolution of 512 × 512 from the in-air dataset Flickr2K. These images were subsequently resized to 256 × 256 and combined with the SUIM dataset to form the input data. Our proposed network was initially trained for 100 epochs with a learning rate of $2 \times e^{-4}$, followed by a

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TABLE I
SEGMENTATION ACCURACY OF DIFFERENT UIE METHODS ON DEEPLABV3+ NETWORK

	BW	HD	PF	WR	RO	RI	FV	SR	mIoU ↑	mPA ↑
Raw	85.0	79.4	45.5	75.4	67.6	73.2	74.9	63.8	70.6	83.8
UDCP [54]	85.8	77.0	45.7	73.0	66.9	73.5	76.1	63.7	70.2	82.7
FGAN [26]	87.1	77.1	43.7	70.7	72.1	74.0	72.8	64.7	70.3	83.6
Water-Net [24]	87.5	77.7	45.3	74.1	70.1	74.5	76.7	66.6	71.6	84.2
Fusion [17]	87.6	77.4	43.6	75.5	70.0	74.6	75.0	65.5	71.1	83.2
UWCNN [30]	87.5	76.5	43.0	70.9	67.3	73.9	75.3	66.5	70.1	82.4
TACL [47]	87.2	76.9	48.3	73.7	67.5	75.1	74.5	67.1	71.3	83.4
U-Trans [33]	85.5	75.4	47.5	77.1	58.1	74.5	74.6	65.8	69.8	83.2
AAC [14]	88.1	78.8	46.9	73.3	65.8	76.6	76.8	69.0	72.0	83.8
Ours	87.7	81.3	44.3	79.6	72.7	76.8	77.2	67.3	73.4	84.7

TABLE II
SEGMENTATION ACCURACY OF DIFFERENT UIE METHODS ON THE U-NET NETWORK

	BW	HD	PF	WR	RO	RI	FV	SR	mIoU ↑	mPA ↑
Raw	87.9	88.4	41.9	73.7	73.6	74.3	80.9	66.0	72.2	82.5
UDCP [54]	87.1	76.0	40.1	68.0	71.2	73.2	77.7	62.6	69.5	81.1
FGAN [26]	86.9	78.6	48.0	73.6	72.3	74.5	78.6	63.3	71.9	79.9
Water-Net [24]	88.2	82.3	47.0	77.1	70.3	74.5	81.3	65.1	73.2	82.8
Fusion [17]	87.3	80.7	43.6	74.9	73.5	75.2	81.9	64.3	72.6	82.4
UWCNN [30]	87.6	81.7	43.2	73.2	70.3	73.9	79.0	64.5	71.7	82.2
TACL [47]	87.5	76.8	48.0	73.6	72.3	74.5	78.6	63.3	71.9	82.2
U-Trans [33]	87.0	81.1	43.1	74.7	78.7	73.5	79.8	64.6	72.8	82.9
AAC [14]	87.2	80.8	42.7	77.5	78.8	74.0	79.2	64.5	73.1	83.1
Ours	88.8	85.4	45.9	77.3	78.2	75.7	81.8	66.1	74.8	83.9

decay to 0 for the subsequent 100 epochs. In addition, a batch size of 1 was employed, and Adam is adopted as the optimizer.

C. Machine Vision Tasks Performance Comparison

To validate the performance of our proposed model in underwater computer vision tasks, we compare it with the state-of-the-art UIE methods on several machine tasks. The comparative methods include two physics-based methods: UDCP [54] and AAC [14]; an image-based method: Fusion [17]; as well as five deep learning-based methods: FGAN [26], UWCNN [30], Water-Net [24], TACL [47], and U-Trans [33]. It is worth noting that TACL specifically considers the impact of enhancement on tasks such as detection. In addition, we compare the enhanced images with the raw images as a baseline for evaluation.

1) Evaluation on Semantic Segmentation: To compare the segmentation performance of our proposed method and the comparison methods, we conduct training on two widely used segmentation networks, namely, DeepLabv3+ [60] and U-Net [61], with the enhanced image and the raw image, respectively. The evaluation includes the assessment of IoU for eight categories (BW, HD, PF, WR, RO, RI, FV, and SR), mIoU, and mPA.

Table I presents the segmentation results achieved with DeepLabv3+, while Table II shows the results obtained with U-Net. The results indicated by the red font represent the top-performing outcomes, and those indicated by the blue font represent the second-best outcomes. Our proposed method consistently outperforms both the raw images and the other comparative algorithms in terms of segmentation quality on both DeepLabv3+ and U-Net. It is worth noting that enhancement methods prioritizing perceptual visual quality often fail to improve segmentation performance. The physics-based method UDCP exhibits the poorest performance on both segmentation models. In contrast, our proposed method demonstrates superior performance and achieves higher segmentation accuracy compared to the raw images, improving mIoU by 2.8% and 2.6% on the two segmentation networks.

To visually showcase the effectiveness of our proposed method, Fig. 6 presents the visual results obtained on the SUIM test set. The images are enhanced using our method and the comparative algorithms and subsequently evaluated using the trained DeepLabv3+ network. The qualitative evaluation demonstrates that our proposed method achieves the results that closely resemble the ground truth, indicating its high fidelity in preserving the underlying semantic information. These findings provide strong evidence of the superiority of our proposed method over existing approaches in enhancing underwater images for improved segmentation performance.

2) Evaluation on Object Detection: To investigate the impact of UIE on object detection task, we apply the enhanced results to multiple detection algorithms, including RetinaNet [62], SSD [63] (one-stage detection network), Faster-RCNN [64] (two-stage detection network), and an underwater object detection network FERNet [65].

The quantitative comparisons of the detection performance on these three networks are presented in Table III. 0B, 1B, and 2B, respectively, correspond to three categories of objects in

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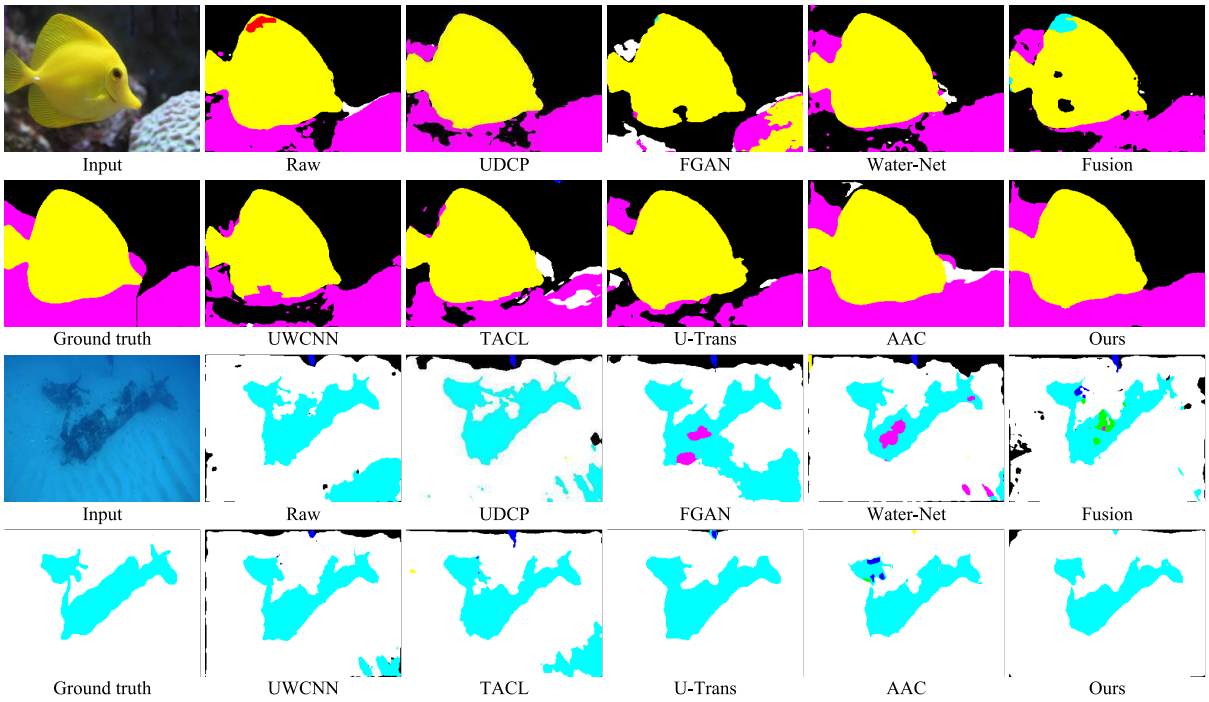


Fig. 6. Visualization results of our proposed method, six comparison algorithms, and raw images on the SUIM test set. We apply the enhanced images and raw images to the DeepLabv3+ model and obtain the corresponding test results. The prediction maps of our method exhibit the closest resemblance to the ground truth, demonstrating the superiority of our approach in underwater semantic segmentation.

TABLE III
OBJECT DETECTION ACCURACY OF DIFFERENT UIE METHODS ON FOUR OBJECT DETECTION NETWORKS

	RetinaNet				SSD				Faster-RCNN				FERNet			
	0B	1B	2B	mAP ↑	0B	1B	2B	mAP ↑	0B	1B	2B	mAP ↑	0B	1B	2B	mAP ↑
Raw	71.2	72.8	74.3	72.8	57.1	58.1	62.6	59.3	65.5	62.8	64.5	64.3	69.8	69.3	71.5	70.2
UDCP [54]	64.2	63.6	67.4	65.1	55.2	56.4	61.8	57.8	63.1	61.1	61.3	61.8	68.2	66.4	68.5	67.7
FGAN [26]	67.8	68.1	70.0	68.6	54.3	56.1	61.3	57.2	63.5	61.1	63.5	62.7	68.8	66.7	69.4	68.4
Water-Net [24]	63.6	66.1	67.5	65.7	56.2	57.0	61.6	58.3	63.9	61.2	63.1	62.7	67.1	65.6	68.0	66.9
Fusion [17]	58.7	61.5	64.9	61.7	56.0	56.8	61.1	58.0	62.4	60.2	62.3	61.6	64.1	63.0	66.4	64.5
UWCNN [30]	64.1	66.0	68.2	66.1	54.4	56.4	61.0	57.3	61.4	59.1	60.8	60.4	67.6	67.5	68.9	68.0
TACL [47]	70.2	75.2	73.1	72.8	56.1	57.2	62.0	58.4	63.4	61.9	63.7	63.0	69.7	70.2	71.3	70.4
U-Trans [33]	63.1	64.2	66.7	64.7	56.1	56.4	60.2	57.6	62.0	60.7	62.6	61.8	67.2	67.0	68.5	66.8
AAC [14]	65.4	64.8	68.4	66.2	56.2	56.8	60.8	57.9	62.2	61.5	63.1	62.3	66.7	67.0	67.6	67.1
Ours	75.3	79.2	77.6	77.4	65.8	63.0	66.3	65.0	68.5	68.5	67.4	68.1	71.5	71.0	72.9	71.8

the ChinaMM dataset: holothurian, echinus, and scallop. Our proposed method significantly improves the mean average precision (mAP) of all four detectors, with improvements of 4.6%, 5.7%, 3.8%, and 1.6% over the raw images. In contrast, the detection performance of other comparative methods shows varying degrees of decline across all three detection networks. Among them, TACL, which jointly trained the detection task, achieves accuracy close to that of the raw images. The other methods do not consider machine vision tasks and their trained or evaluated data characteristics differ significantly from those of the ChinaMM dataset, resulting in more severe degradation in detection performance.

Fig. 7 provides a visual comparison of the detection results on RetinaNet. From the visualizations, it can be observed that the detection on the raw images and the enhanced images using the comparative methods are prone to missing objects in challenging areas, where the object is small and blurred and the object is incompletely occluded. In addition, our proposed method demonstrates higher confidence in the detected objects. Even though some methods appear to be visually superior, such as Water-Net and TACL, they do not exhibit a direct correlation with the object detection task.

3) Evaluation on Saliency Detection: We further conduct underwater saliency detection task on the enhanced images using the U2NET [66] network.

Table IV presents the comparison results on two datasets in terms of four evaluation metrics: F-measure, S-measure, E-measure, and MAE. Our proposed method consistently outperforms the other methods in all four metrics, indicating its superior performance in underwater saliency detection. Conversely, the majority of comparative methods fall short compared to the raw images, while our method achieves improvements of 3.4% in F-measure, 0.8% in S-measure, and 2.0% in E-measure, as well as a reduction of 0.008 in

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TABLE IV
SALIENCY DETECTION ACCURACY OF DIFFERENT UIE METHODS ON THE U2NET NETWORK

	UFO-120 Dataset				USOD10K Dataset			
	F-measure ↑	S-measure ↑	E-measure ↑	MAE ↓	F-measure ↑	S-measure ↑	E-measure ↑	MAE ↓
Raw	71.3	78.3	78.6	0.133	73.5	85.9	85.5	0.055
UDCP [54]	70.0	76.2	76.7	0.152	70.7	83.6	83.5	0.063
FGAN [26]	71.4	77.0	77.7	0.145	72.3	84.9	84.4	0.059
Water-Net [24]	71.9	77.8	78.6	0.138	70.3	85.0	83.7	0.062
Fusion [17]	71.4	78.4	77.5	0.136	74.5	85.0	86.3	0.058
UWCNN [30]	71.0	77.4	77.8	0.141	72.5	86.4	85.0	0.053
TACL [47]	67.5	75.6	72.9	0.161	73.0	85.5	84.8	0.058
U-Trans [33]	70.0	78.4	77.8	0.135	72.9	85.7	85.2	0.056
AAC [14]	72.2	77.5	79.2	0.148	70.5	84.7	83.5	0.064
Ours	74.7	79.1	80.6	0.125	76.0	86.7	87.2	0.049

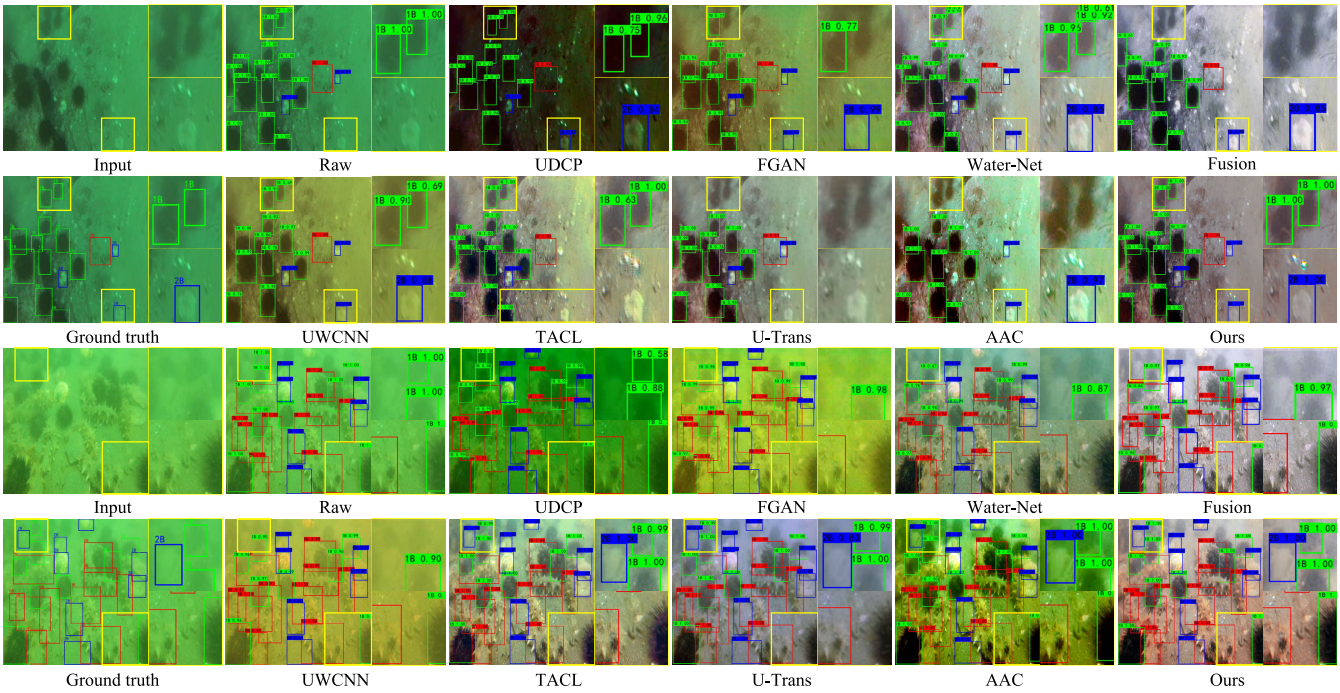


Fig. 7. Visualization results of our proposed method, six comparison algorithms, and raw images on the ChinaMM test set. We apply the enhanced images and raw images to the RetinaNet model and obtain the corresponding test results. 0B indicates the detected holothurian, 1B indicates the echinus, and 2B indicates the scallop. The yellow boxes show some hard areas that are difficult to detect. It can be seen that our method also has a high detection rate in these hard areas.

MAE on the UFO-120 dataset, and 2.5% in F-measure, 0.8% in S-measure, and 1.7% in E-measure, as well as a reduction of 0.006 in MAE on the USOD10K dataset. Although TACL performs better than other comparative algorithms in object detection, it shows the poorest performance in saliency detection on the UFO-120 dataset. This indicates its lack of generalization to other underwater machine vision tasks. In contrast, our method, which incorporated segmentation-related information during training, demonstrated its advantages in object detection and saliency detection. This further confirms the strong generalization ability of our approach.

To provide a qualitative assessment, Figs. 8 and 9 present the visualizations of the saliency detection results. It is evident that methods, such as UDCP and Fusion, incorrectly identify the background as salient objects. In contrast, our proposed enhancement method, when compared to UWCNN, exhibits more accurate edge detection and, subjectively, even outperforms the ground truth. These results demonstrate the effectiveness of our proposed enhancement method in improving the performance of underwater saliency detection. It outperforms the comparative methods in terms of various evaluation metrics and yields more accurate and visually appealing saliency detection results.

D. Ablation Study

To validate the contributions of each component, we conduct several ablation experiments. Three factors are considered, including the DR, the SACM, and the DTGB. Model A represents the baseline without the addition of DR, SACM, and DTGB. Model B represents the baseline with the inclusion of

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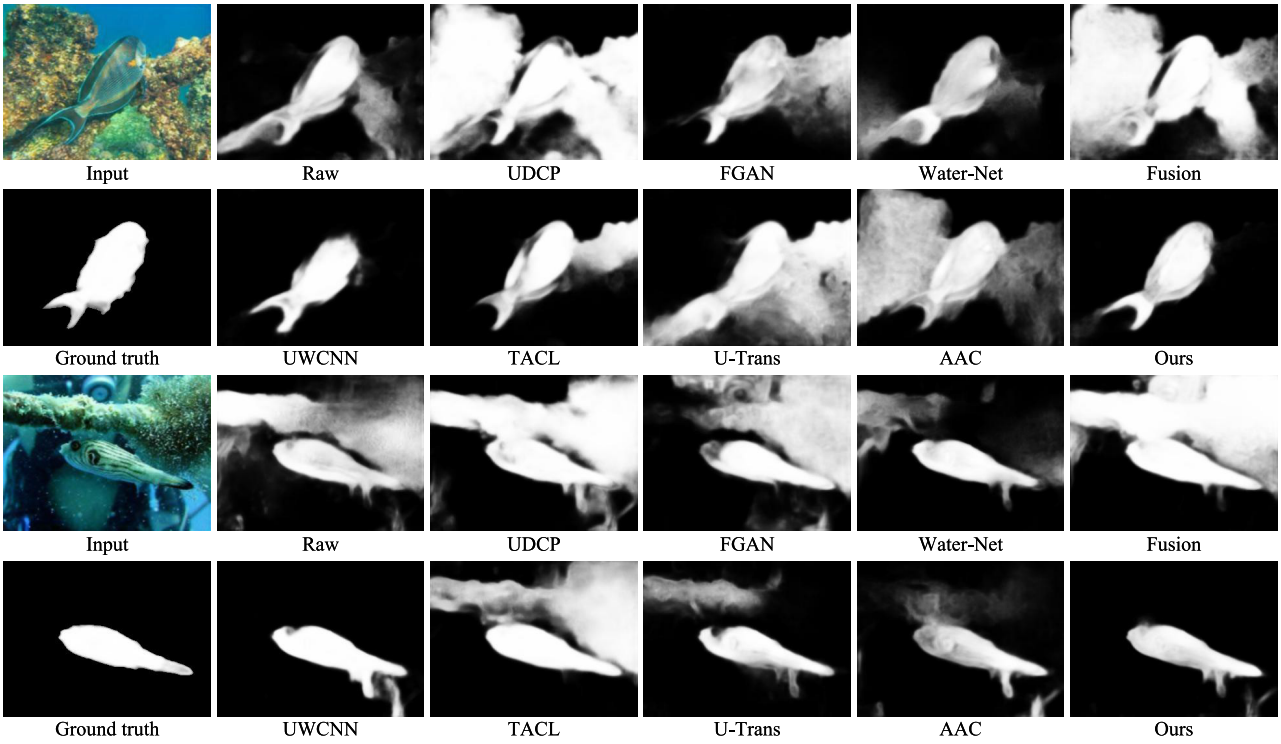


Fig. 8. Visualization results of our proposed method, six comparison algorithms, and raw images on the UFO-120 test set. We apply the enhanced images and raw images to the U2NET model and obtain the test results. Our method detects salient objects more accurately and produces smoother edges.

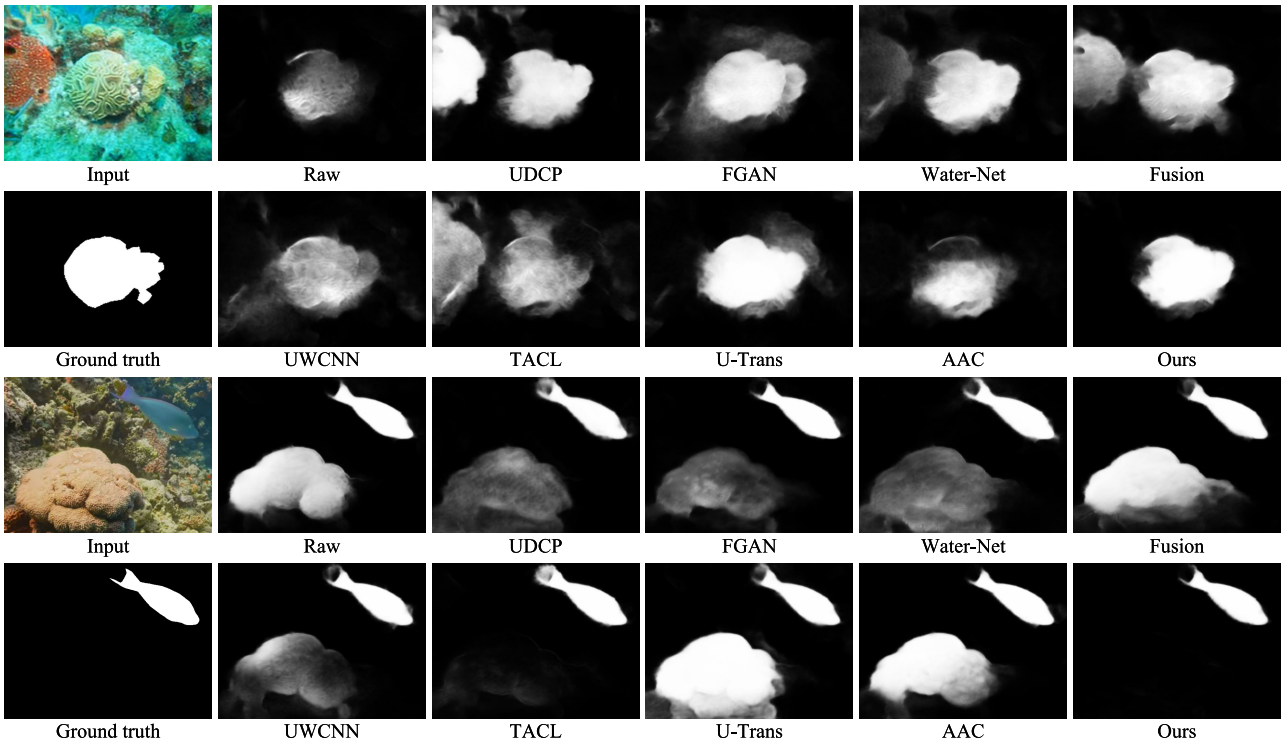


Fig. 9. Visualization results of our proposed method, six comparison algorithms, and raw images on the USOD10K test set. We apply the enhanced images and raw images to the U2NET model and obtain the corresponding test results. Our method detects salient objects more accurately and produces smoother edges.

DR. Model C represents the baseline with the inclusion of DR and DTGB. Model D represents the baseline with the inclusion of DR and SACM.

1) Study on Semantic Segmentation: The results on the semantic segmentation task are presented in Table V. Compared to baseline (Model A), the addition of the DR (Model B)

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TABLE V
ABLATION EXPERIMENTS ON SEMANTIC SEGMENTATION

	DR	SACM	DTGB	DeepLabv3+		U-Net	
				mIoU	mPA	mIoU	mPA
Model A	✗	✗	✗	65.9	82.3	70.2	80.3
Model B	✓	✗	✗	70.7	82.7	71.6	81.5
Model C	✓	✗	✓	72.7	84.1	73.0	82.7
Model D	✓	✓	✗	71.8	83.6	72.1	82.3
Ours	✓	✓	✓	73.4	84.7	74.8	83.9

TABLE VI
ABLATION EXPERIMENTS ON OBJECT DETECTION EVALUATED ON MAP

	DR	SACM	DTGB	RetinaNet	SSD	Faster-RCNN	FERNet
Model A	✗	✗	✗	62.1	55.4	59.3	65.2
Model B	✓	✗	✗	66.8	57.0	62.4	67.0
Model C	✓	✗	✓	74.7	61.4	65.6	70.1
Model D	✓	✓	✗	74.0	62.2	65.2	69.8
Ours	✓	✓	✓	77.4	65.0	68.1	71.8

TABLE VII
ABLATION EXPERIMENTS ON SALIENCY DETECTION TASK ON UFO-120

	DR	SACM	DTGB	F-measure	S-measure	E-measure	MAE ↓
Model A	✗	✗	✗	67.2	75.1	74.4	0.145
Model B	✓	✗	✗	69.3	76.7	76.8	0.136
Model C	✓	✗	✓	72.4	78.2	80.0	0.131
Model D	✓	✓	✗	73.3	78.8	79.7	0.128
Ours	✓	✓	✓	74.7	79.1	80.6	0.125

TABLE VIII
ABLATION EXPERIMENTS ON SALIENCY DETECTION TASK ON USOD10K

	DR	SACM	DTGB	F-measure	S-measure	E-measure	MAE ↓
Model A	✗	✗	✗	72.8	84.8	84.2	0.058
Model B	✓	✗	✗	73.7	85.5	85.4	0.056
Model C	✓	✗	✓	74.3	86.1	85.8	0.053
Model D	✓	✓	✗	74.0	86.4	85.6	0.053
Ours	✓	✓	✓	76.0	86.7	87.2	0.049

improves the mIoU by 4.8% and 1.4% on DeepLabv3+ and U-Net, respectively. Compared to Model B, the addition of the DTGB (Model C) contributes to a performance improvement in mIoU by 2.0% and 2.4%, and the addition of the SACM (Model D) contributes to a performance improvement by 1.1% and 0.5%.

2) Study on Object Detection: The results on the object detection task are shown in Table VI. Compared to baseline (Model A), the addition of the DR (Model B) improves the mAP by 1.6%–4.7% on four detection networks. Compared to Model B, the addition of the DTGB (Model C) contributes to a performance improvement in mAP by 3.1%–7.9%, and the addition of the SACM (Model D) contributes to a performance improvement by 2.8%–7.2%.

3) Study on Saliency Detection: The results on the saliency detection task are presented in Table VII. Compared to baseline (Model A), the addition of the DR (Model B) improves the F-measure by 2.1% and 0.9% on two datasets. Compared to Model B, the addition of the DTGB (Model C) contributes to a performance improvement in F-measure by 3.1% and 0.6%, and the addition of the SACM (Model D) contributes to a performance improvement by 4.0% and 0.3%.

These results demonstrate the effectiveness of the DR, SACM, and DTGB across multiple tasks, including semantic segmentation, object detection, and saliency detection. The improvements achieved in performance metrics validate the contributions of our proposed approaches.

V. CONCLUSION

In this article, we have proposed an unsupervised UIE framework based on DR, specifically designed for machine vision tasks. By disentangling the underwater image into task-friendly content features and task-unfriendly distortion features, and incorporating a semantic-aware feature extractor and contrastive loss, we effectively preserve the semantic content while mitigating underwater distortions. The integration of a DTGB further improves performance in underwater machine vision tasks. Experimental results demonstrate the superiority of our approach compared to state-of-the-art methods on a variety of underwater machine vision tasks, highlighting its potential for enhancing machine vision performance in challenging underwater scenarios. Moreover, our proposed method can be extended to various data types, such as remote sensing images and hazy images, and can better support research and applications in the field of geoscience.

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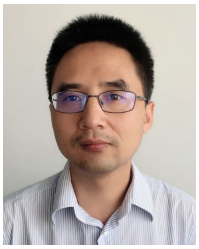
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